2010376103 CSE4261 Assignment-3

Omor Niloy

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1 Feature Extraction Power of CNN Before and After Transfer Learning

The feature extraction power of a CNN refers to how well it can represent input images in a way that separates different classes. In this case, a ResNet-50 model pretrained on ImageNet is used as the base CNN. Before transfer learning, the model is not yet adapted to MNIST, so the features are not very class-specific. After transfer learning, the CNN becomes much better at capturing patterns in MNIST digits, leading to clearer separation between classes in visualizations.

2 Dataset

The MNIST dataset contains 70,000 grayscale 28×28 images of handwritten digits (0–9). For visualization speed, 1,000 training samples were used. Each image was converted to 3 channels by stacking the single channel, resized to 224×224, and preprocessed with `preprocess_input()` from the Keras ImageNet utilities. Labels were one-hot encoded into 10 classes.

3 Experimental Setup

A ResNet-50 model pretrained on ImageNet was used as the backbone. Two conditions were compared:

1. **Before transfer learning:** ResNet-50 with ImageNet weights. Convolutional base kept frozen. Features extracted from the final global average pooling layer (2048-d).
2. **After transfer learning:** The same ResNet-50 fine-tuned on MNIST. Top classifier replaced with:
   - GlobalAveragePooling2D
   - Dense(256, ReLU)
   - Dropout(0.5)
   - Dense(10, Softmax)

   Training used Adam, categorical cross-entropy, and early stopping. Initially only the new head was trained, then selected deeper layers were unfrozen for fine-tuning.

High-dimensional feature vectors (2048-d) were projected to 2D using three techniques: PCA, t-SNE, and UMAP. The same 1,000 samples were used for both before/after visualizations. Figures 1, 2 were exported.

4 Results

4.1 Before transfer learning

PCA shows heavy class overlap. t-SNE produces some local grouping but classes are mixed. UMAP yields limited cluster structure. Overall separability is weak.

- **PCA:** Features are mixed and overlapping. classes are not well separated.

- **t-SNE:** Some clustering is visible, but many classes are still close to each other.
- **UMAP:** Slightly better than PCA. Some clusters are forming but still not very distinct.

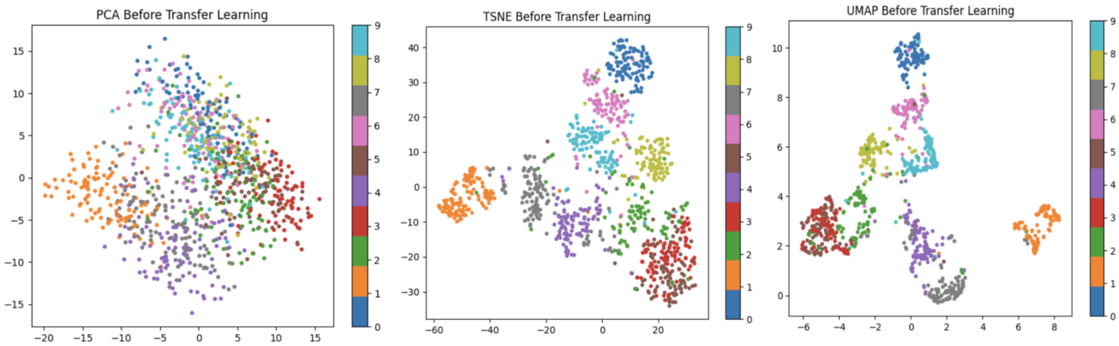

Figure 1: 2D feature visualization before transfer learning (ImageNet only). PCA, t-SNE, and UMAP projections are shown.

4.2 After transfer learning

PCA shows clearer class separation. t-SNE produces tight, well-separated clusters per digit. UMAP also shows compact clusters with minimal overlap. Feature representations become far more discriminative after fine-tuning.

- **PCA:** Classes are more clearly separated than before.
- **t-SNE:** Strong and tight clustering. Each digit class is well grouped.
- **UMAP:** Very clear class separation, showing strong feature learning for MNIST.

Figure 2: 2D feature visualization after transfer learning (fine-tuned on MNIST). PCA, t-SNE, and UMAP projections are shown.

5 Conclusion

Transfer learning markedly improves ResNet-50's feature extraction on MNIST. Pretrained ImageNet features provide a starting point, but fine-tuning yields compact, class-separable embeddings in 2D projections. The effect is visible across PCA, t-SNE, and UMAP.

6 Code

Github code link